

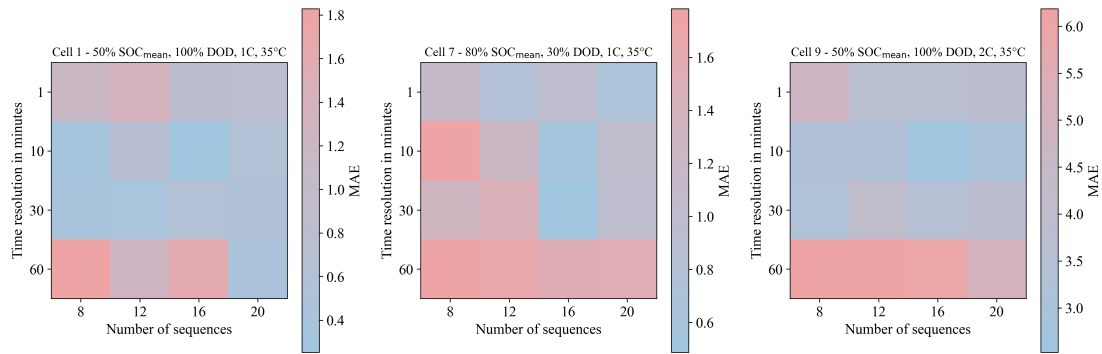


Figure 5.5: Matrix of MAE values shown in percentage points of SOH on the 0 to 100% SOH scale. The corresponding normalized MAE on a 0 to 1 SOH scale is obtained by scaling the displayed values accordingly. The colour scale indicates the magnitude of the prediction error and highlights that the most favourable operating region is found at intermediate temporal aggregation rather than at the shortest or longest tested interval.

Figure 5.6 compares the predicted and measured SOH trajectories. Cells 7 and 1 represent typical ageing behaviour and are reproduced well. The prediction for Cell 9 follows parts of its pronounced U-shaped progression, but the estimated SOH values show larger pointwise and systematic deviations from the capacity-based reference. Its atypical shape does not itself produce a high error. If that shape were reproduced quantitatively, the points would remain close to the identity line in Figure 5.7 and both MAE and MSE would be small. Cell 9 instead provides an example in which the model captures part of the qualitative development but generalises less accurately in absolute SOH.

Figure 5.7 complements the trajectory-based view with a direct comparison between predicted and measured SOH values. The distance of each point from the identity line is the individual prediction residual. MAE summarizes the average absolute magnitude of these residuals, whereas MSE gives greater weight to the larger deviations. For the favourable individual cells, the reported best errors are MSE values of $0.410\%^2$ and $0.118\%^2$ and MAE values of 0.487% and 0.253% . These correspond to normalized MAE values of 0.00487 and 0.00253 on a 0 to 1 SOH scale. For Cell 9, the MAE increases to 2.528% and the MSE to $8.327\%^2$, which quantifies the reduced prediction accuracy visible in the trajectory and scatter plots. Across the evaluated cases, the average MAE is 1.10% , corresponding to a normalized MAE of 0.0110 , and the average MSE is $2.95\%^2$, corresponding to a normalized MSE of $2.95 \cdot 10^{-4}$. The prediction plots additionally show that